

HTML and CSS: Comprehensive Guide



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HTML (HyperText Markup Language) and CSS (Cascading Style Sheets) are the foundation of web development. HTML provides the structure of a webpage, while CSS styles and layouts the content. This guide will teach you the basics of HTML and CSS, with examples, exercises, and quizzes to reinforce learning.

1. Introduction to HTML

HTML is the backbone of any webpage, defining its structure with elements (tags).

Example: Basic HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>My First Webpage</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is a paragraph with some content.</p>
</body>
</html>
```

Common HTML Tags:

Tag	Description
<h1> to <h6>	Headings (h1 is the largest, h6 is the smallest).
<p>	Paragraph text.
<a>	Hyperlink.
	Image.
,	Unordered and ordered lists.

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`<table>` Table.

`<div>` Block-level container.

`<span>` Inline container.

Example: Image and Link

```

<a href="https://example.com" target="_blank">Visit Example</a>
```

2. Introduction to CSS

CSS styles the structure provided by HTML. It defines colors, layouts, fonts, and much more.

Example: Adding CSS to HTML

Inline CSS:

```
<p style="color: blue; font-size: 20px;">This is a blue paragraph.</p>
```

Internal CSS:

```
<head>
  <style>
    body {
      background-color: #f0f0f0;
    }
    h1 {
      color: green;
    }
  </style>
</head>
```

External CSS:

```
<head>
  <link rel="stylesheet" href="styles.css">
</head>
```

styles.css:

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```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 0;
}
h1 {
  color: navy;
}
```

3. Box Model

The CSS Box Model describes the layout of elements, including:

1. **Content:** The content inside the element.
2. **Padding:** Space between content and border.
3. **Border:** Surrounds the padding.
4. **Margin:** Space outside the border.

Example:

```
<div style="width: 200px; padding: 10px; border: 2px solid black;
margin: 20px;">
  Box Model Example
</div>
```

4. CSS Selectors

Selectors specify which HTML elements a CSS rule applies to.

Common Selectors:

Selector	Example	Description
Universal	*	Selects all elements.
Type	p	Selects all <p> elements.
Class	.className	Selects elements with a class.
ID	#idName	Selects an element with an ID.

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Descendant `div p` Selects `<p>` inside `<div>`.
t

Example:

```
<div class="container" id="main">
  <p class="text">This is a paragraph.</p>
</div>
```

CSS:

```
#main {
  background-color: lightgray;
}
.text {
  color: blue;
  font-size: 18px;
}
```

5. Responsive Design

Responsive design ensures that a website looks good on all devices.

Example: Media Queries

```
body {
  font-size: 16px;
}
@media (max-width: 600px) {
  body {
    font-size: 14px;
  }
}
```

Detailed Examples

Example 1: Creating a Simple Webpage

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

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```

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Simple Webpage</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f9f9f9;
        }
        header {
            background-color: #333;
            color: white;
            padding: 10px 0;
            text-align: center;
        }
        main {
            padding: 20px;
        }
        footer {
            background-color: #333;
            color: white;
            text-align: center;
            padding: 10px 0;
        }
    </style>
</head>
<body>
    <header>
        <h1>Welcome to My Webpage</h1>
    </header>
    <main>
        <h2>About Me</h2>
        <p>This is a simple webpage created using HTML and CSS.</p>
    </main>
    <footer>
        <p>&copy; 2024 My Website</p>

```

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```
</footer>
</body>
</html>
```

Example 2: Creating a Navigation Menu

```
<nav>
  <ul style="list-style: none; padding: 0; display: flex; background:
#333;">
    <li style="margin-right: 20px;"><a href="#" style="color: white;
text-decoration: none;">Home</a></li>
    <li style="margin-right: 20px;"><a href="#" style="color: white;
text-decoration: none;">About</a></li>
    <li><a href="#" style="color: white; text-decoration:
none;">Contact</a></li>
  </ul>
</nav>
```

Exercises

Exercise 1: Build a Portfolio Page

- Create sections for "Home," "Portfolio," and "Contact."
- Style them with internal CSS.

Exercise 2: Style a Table

- Create a table and style it using CSS.

Solution:

```
<table style="width: 100%; border-collapse: collapse;">
  <tr style="background: #f4f4f4;">
    <th style="border: 1px solid #ddd; padding: 8px;">Name</th>
    <th style="border: 1px solid #ddd; padding: 8px;">Age</th>
  </tr>
  <tr>
    <td style="border: 1px solid #ddd; padding: 8px;">Alice</td>
    <td style="border: 1px solid #ddd; padding: 8px;">25</td>
  </tr>
</table>
```

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Exercise 3: Responsive Design

- Add a media query to change the background color for screens smaller than 600px.

Multiple-Choice Questions

Question 1:

What does the <p> tag represent in HTML?

1. Paragraph
2. Picture
3. Page
4. Placeholder

Answer: 1. Paragraph

Question 2:

Which CSS property changes the background color of an element?

1. color
2. background-color
3. font-color
4. text-background

Answer: 2. background-color

Question 3:

What is the purpose of @media in CSS?

1. To add animations.
2. To create media queries for responsive design.
3. To import external CSS files.
4. To style media elements like videos.

Answer: 2. To create media queries for responsive design.

Best Practices

1. **Use Semantic HTML:** Use tags like <header>, <main>, <footer> for better accessibility and SEO.
2. **Keep CSS Organized:** Use external stylesheets for maintainable code.

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3. **Responsive Design:** Test on multiple devices and use media queries.
4. **Consistent Styling:** Use CSS variables to maintain consistent colors and spacing.